DO NOW — RECALL

Magnetic, optical and flash storage

Name: _____ Class: _____ Date: _____

1. Why could a GPU be useful for many independent pixel operations?

2. What does ACC hold?

3. Define non-volatile.

4. Evaluate 11010110_2 AND 00001111_2, then write the resulting byte in hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It can execute many similar operations concurrently.

2. Model answer

Intermediate arithmetic/logical result.

3. Model answer

Data remains stored without power.

4. Model answer

$00000110_2 = 06_{16}$.

Vocabulary — know these terms

magnetic — Bits stored through magnetic states.

optical — Bits encoded/read through optical disc patterns.

flash — Non-volatile electronic storage.

seek latency — Delay in finding data.

backup — Separate recoverable copy of data.

NEW LEARNING

Read, annotate and explain

Identify the storage mechanism

HDDs use magnetised regions on rotating platters; DVD-style media are read optically.

Explain solid-state storage

Flash cells are non-volatile electronic storage without rotating platters or mechanical head seeks.

Evaluate rather than rank universally

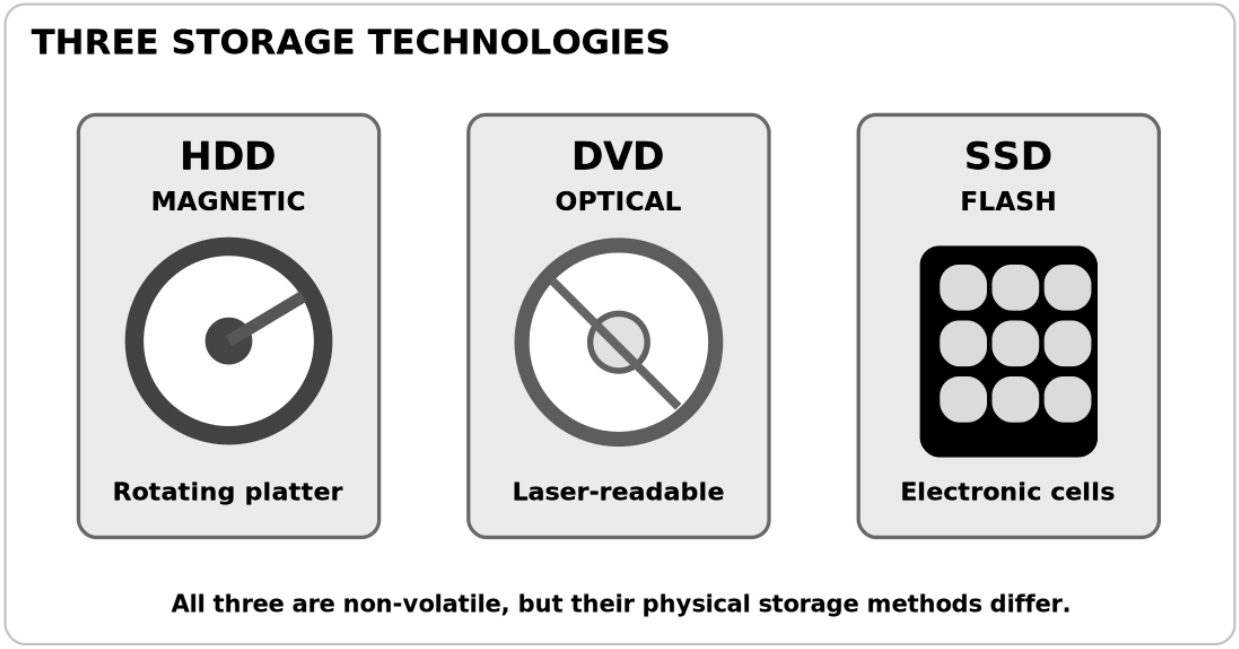
SSD generally has low seek latency and resists mechanical shock; HDD may offer lower cost per GB at large capacities.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

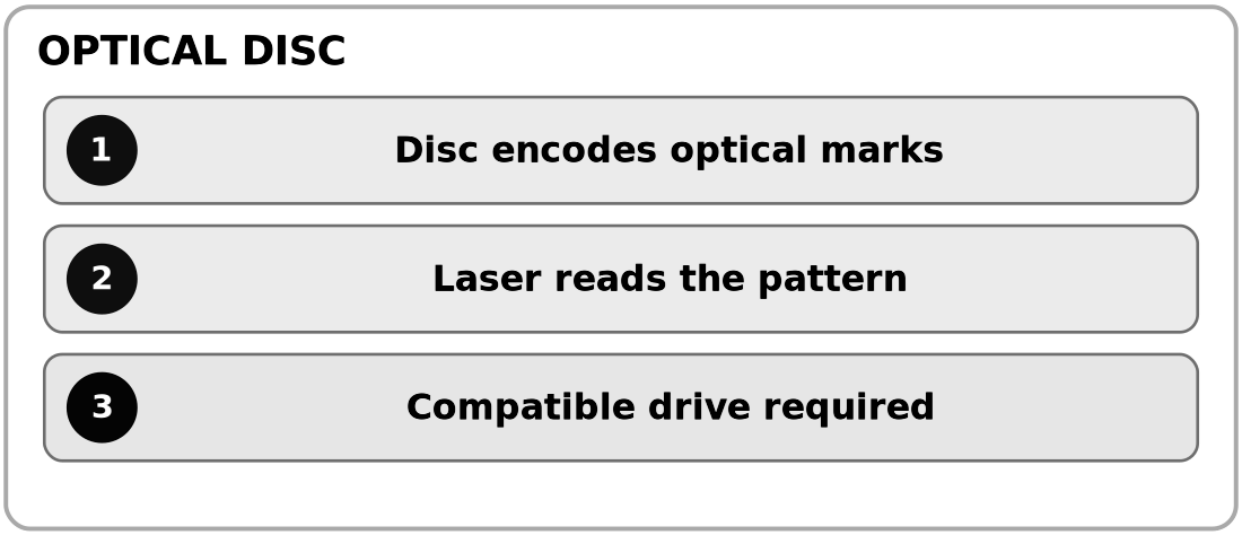
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Magnetic HDD with platter, optical disc with laser-readable marks, SSD with flash memory cells

New Learning

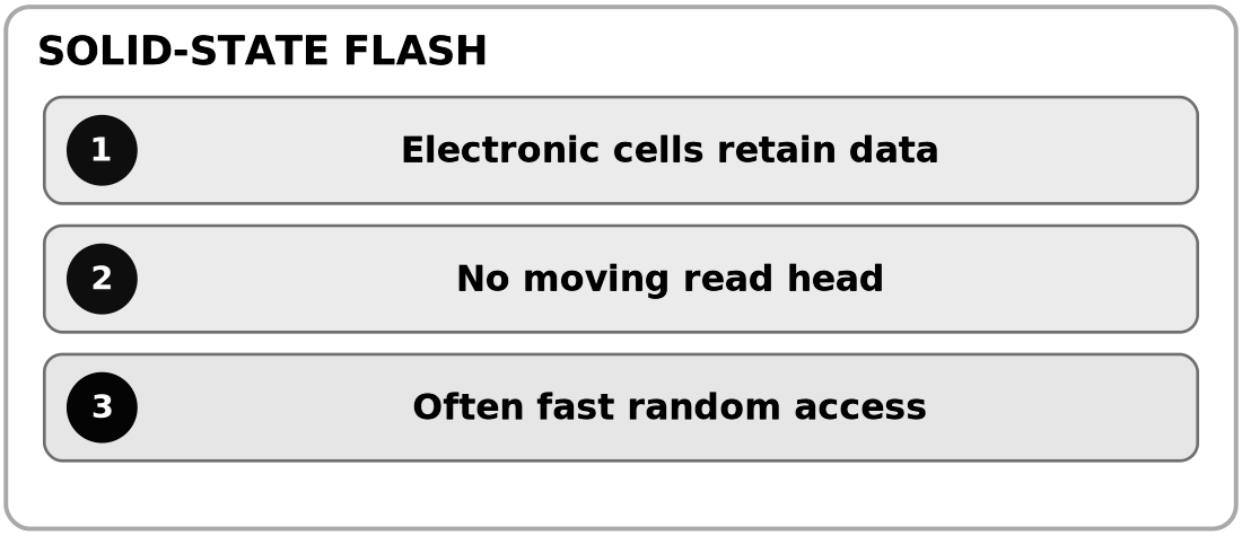


OPTICAL DISC: Disc encodes optical marks; Laser reads the pattern; Compatible drive required

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SOLID-STATE FLASH: Electronic cells retain data; No moving read head; Often fast random access

I DO — WORKED EXAMPLE

Follow the worked method

Choose SSD or HDD for a photographer moving large images between sites.

Predict

An SSD is attractive for frequent portable access and avoiding moving parts.

Explain

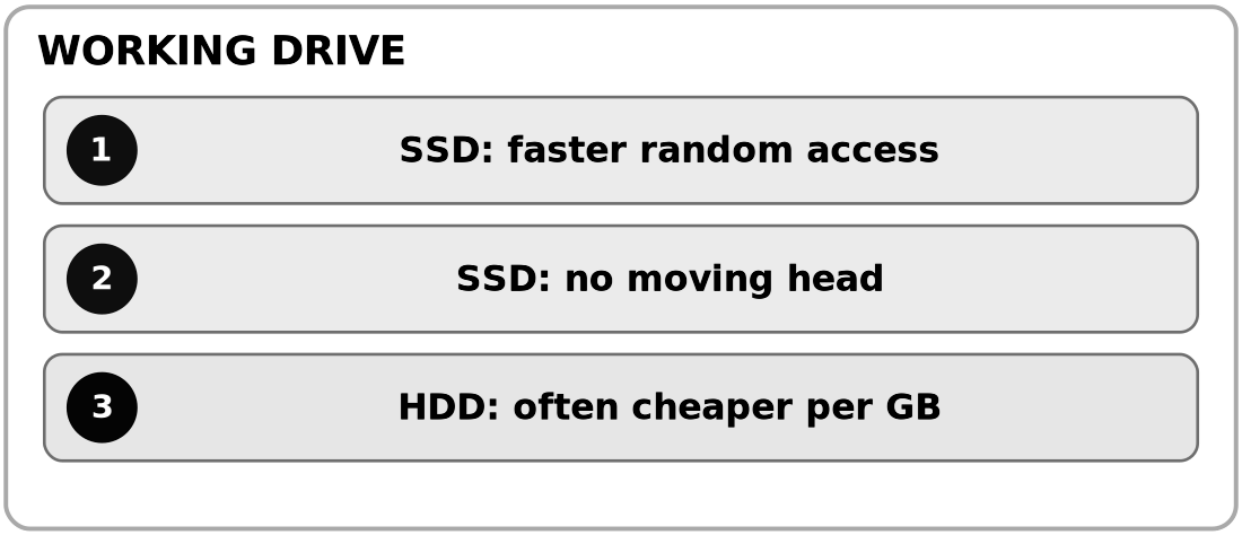
An HDD can offer capacity at a lower cost; backups remain necessary regardless.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose SSD or HDD for a photographer moving large images between sites.

An SSD is attractive for frequent portable access and avoiding moving parts.



WORKING DRIVE: SSD: faster random access; SSD: no moving head; HDD: often cheaper per GB

WE DO — GUIDED PRACTICE

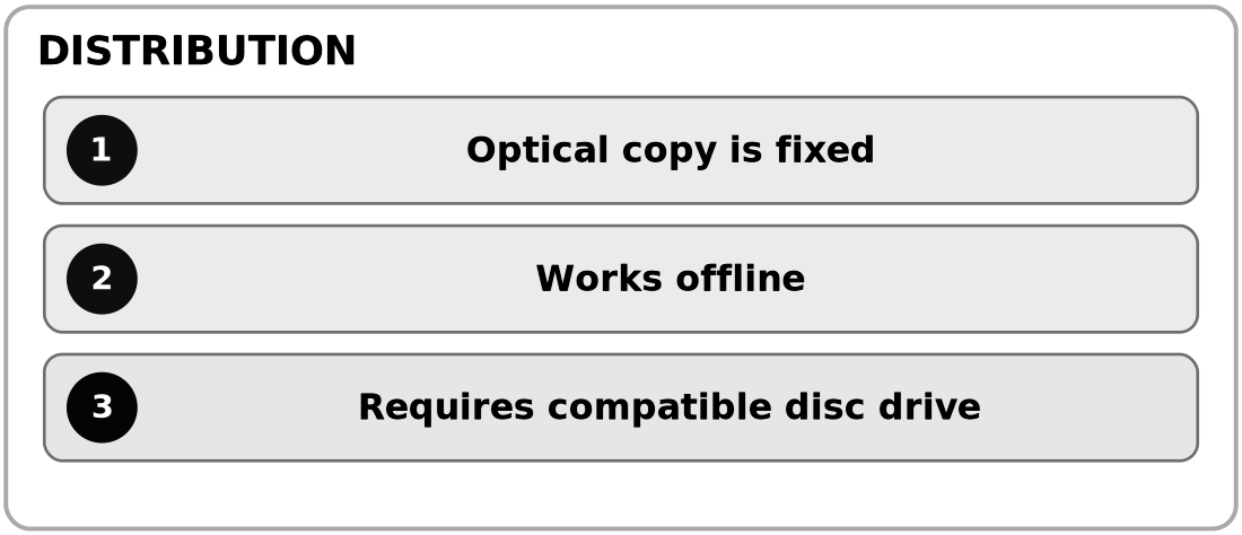
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school distributes a fixed offline exhibition to computers with disc drives. Would an optical disc work?

An optical disc can distribute static content without needing a network connection.



DISTRIBUTION: Optical copy is fixed; Works offline; Requires compatible disc drive

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict An optical disc can distribute static content without needing a network connection.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give storage technology and one physical feature for HDD, DVD and USB flash.

Question 2

Explain two benefits and one drawback of an SSD rather than an HDD for frequent portable editing.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Propose fast working storage and bulk archival storage for a school, explaining backup strategy.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

USB flash and HDDs use the same magnetic storage method.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

Check disc and drive compatibility and whether updates/reuse are needed.

IMPROVE: Explain one correction to your own reasoning.